July 24, 2003

Mr. Larry Campbell, Director of Utilities
City of Cadillac
200 Lake Street
Cadillac, Michigan 49601

Dear Mr. Campbell:

SUBJECT: National Pollutant Discharge Elimination System (NPDES)
Permit No. MI0020257
Designated Name: Cadillac WWTP

Thank you for the timely comments on the Cadillac WWTP draft NPDES permit. My responses to your letter dated July 21, 2003, and Faxed July 23, 2003, are as follows:

Item 1: Permit Length

Permits for your basin year were received April 1, 2003. Your previous permit required application submittal on April 1, 2002. Your review will now be in the proper year so separate studies do not have to be done, reducing costs to the taxpayers of Michigan. Also, the discharge has potential to cause exceedance of water quality standards for mercury and acute and chronic toxicity. These are expensive tests. Do you really want to sample these parameters for five years, rather than 3 and a half years.

We have minimized your sampling, while still responsibly providing you discharge limits that will protect the public. Whether the permit is 3 or 5 years, the effective date and initiation of limits will stay the same.

Item 2: Total Residual Chlorine

NPDES permits are based on the application filed for the facility. The application did not provide a date that UV would be used, or a description of your proposed disinfection system. In the future, please provide this information as soon as it is available.

As for the limit, it will remain. You should have no problem meeting the limit if you use no chlorine. Monitoring frequency reduction can be allowed through the district office. I will add language to allow this to your permit.

Item 3: Lindane

Based on stream flow and discharge characteristics provided with your application (and staff sampling), there is high potential to exceed water quality standards for this parameter. A limit is required, with associated weekly monitoring, after a monitoring period to allow you to reduce Lindane, comply with the limit, and hopefully show no detection in samples so we can remove the limit when the next permit is issued. This is another reason I would suggest a shorter term permit.

The draft presently proposes a one year monitoring period. If you are not detecting any Lindane, and all sources are removed, we need to see the data and quantification level. How long will it take you to investigate and remove Lindane sources if a year is not enough time?

Provide the basis for your time estimate. We use default time factors in drafting permits and can often provide more time if it is justified.

Item 4: Total Mercury

Based on stream flow and discharge characteristics provided with your application (and staff sampling), there is high potential to exceed water quality standards for this parameter. A limit is required, with associated weekly monitoring, after a one year monitoring period to allow you to reduce mercury sources to the Lowest Concentration Achievable. The LCA, which is presently 30 Ng/l, may be lowered to 10 Ng/l the first of the year since many facilities are operating at that level. Both LCA's are variances allowed so dischargers make reasonable forward progress toward meeting the water quality standard of 1.3 Ng/l.

The minimization program is good. We recognize this is a challenging parameter and try to give you some flexibility in designing your own program, while at the same time providing both you and the public limits for the discharge that are protective [of water users].

Item 5: WET Testing

Monthly testing is required to assess compliance with a WET limit. In other words, when the limit is in effect, you must test monthly. I can reduce the monitoring phase of the permit to quarterly. However, be aware that the number of data samples and toxic units measured in the discharge are extremely important in determining if we can remove these requirements in your next permit. Too few samples and a couple of high measurements and the limits will continue (with no adjustment monitoring period) in the next permit.

WET testing is also a real world indicator of toxicity problems. While we provide limits for some parameters, combinations of parameters or unreported parameters may cause toxicity that these tests uncover. This is why WET tests are so important to assess complex municipal treatment plant discharges. WET tests are a Federal requirement.

Item 6: Dissolved Oxygen

The proposed Dissolved Oxygen limits have not changed. In your current permit the Dissolved Oxygen, CBOD5, and ammonia limits are based on meeting the cold water dissolved oxygen standard of 5 mg/l in the Clam River. These parameters are interrelated. In cold weather (December through April) when water is more easily saturated with oxygen and river drought flows are higher, we allowed discharge at 5.0 mg/l. Higher ammonia limits and CBOD5 limits were allowed during this time also based on the difficulty of ammonia treatment in cold weather and lower contact uses of the River.

During the warm weather months drought flows are lower, ammonia treatment in the plant is more effective, and river use (fishing, swimming) requires a higher level of protection. This is the period you operate at peak efficiency and discharge stable effluent (CBOD of 10/4 mg/l and Ammonia Nitrogen of 2/0.5 mg/l). The Dissolved Oxygen standard does not change, but the discharge combination the discharge must meet does. I am sure previous staff worked with your staff to pick the most beneficial combination that meets standards. Perhaps you were involved. If you need a different combination, please request this modeling when you reapply in April of 2007.

Item 7: Additional Monitoring

Additional monitoring requirements were proposed and promulgated in the Federal Register in 40 CFR Part 122.2(j) as explained in our informational letter dated March 24, 2000 (enclosed). POTWs exceeding 1 MGD design capacity or with a Federal Industrial Pretreatment Program or required to develop a FIPP, or otherwise required by the permitting authority to provide the information are required to submit this additional information. The additional information: three scans, including a low level mercury test, and eight WET tests (four on each of two species) are required for each permit reapplication.

We agree that monitoring should be minimized while still adequately protecting the environment. Many of the above requirements were mandated by the legislature and the EPA. We have worked within the statutes and your application to make the new requirements as reasonable as possible.

Sincerely,

Diane Carlson, P.E.
Lakes Michigan and Superior Permits Unit
Permits Section
Water Division
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Enclosure:
cc: Mr. Jeffrey Dietlin, Water Resources Supervisor
Cadillac District Office, Water Division, DEQ
Ms. Yuronda Glasscoe, SWQAS, Water Division, DEQ
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